

# Pramod Patil

## ResearchPaper (1)

 RBL 2025-26

---

### Document Details

Submission ID

trn:oid:::3618:136916296

Submission Date

Apr 28, 2026, 4:19 PM GMT+5:30

Download Date

Apr 28, 2026, 4:38 PM GMT+5:30

File Name

ResearchPaper (1).pdf

File Size

156.1 KB

5 Pages

3,671 Words

21,747 Characters

## 22% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

### Caution: Review required

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

## Detection Groups



33 AI-generated only 22%

Likely AI-generated text from a large-language model.



0 AI-generated text that was AI-paraphrased 0%

Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

### Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

## Frequently Asked Questions

### How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (\*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

### What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



# AuraID: A Non-Blockchain Zero-Knowledge Proof System for Privacy-Preserving Student Identity Verification

Neev Jain  
Thakur College of  
Engineering and Tech.  
Mumbai, Maharashtra  
1032230132@tcetmumbai.in

Ankush Sahu  
Thakur College of  
Engineering and Tech.  
Mumbai, Maharashtra  
1032230146@tcetmumbai.in

Pratyush Pandey  
Thakur College of  
Engineering and Tech.  
Mumbai, Maharashtra  
1032230135@tcetmumbai.in

**Abstract**—In many institutions today, student identity verification still follows a basic approach where showing proof for one detail, such as age or enrollment, requires revealing the entire identity document. This often exposes unnecessary personal information. To address this issue, this paper introduces AuraID, a decentralized identity system that does not rely on blockchain. It uses Zero-Knowledge Proofs (ZKP) to allow students to verify specific details without sharing their complete identity.

The system is built using a Spring Boot backend for credential issuance and an Angular-based web wallet that securely stores data and generates proofs directly in the browser using WebAssembly. In addition, AuraID includes an AI-based anomaly detection system to identify suspicious verification activities and an OCR module to convert physical ID cards into digital credentials.

Experimental results show that proof generation takes less than 120 milliseconds, while verification takes under 30 milliseconds on standard hardware. Overall, AuraID provides a faster and more privacy-focused alternative to traditional and blockchain-based identity systems, with potential for future improvements using federated learning and privacy-preserving AI techniques.

**Index Terms**—Zero-Knowledge Proofs, Decentralized Identity, Angular, Spring Boot, Non-Blockchain dID, Selective Disclosure, Privacy-Preserving AI, Federated Learning, WebAssembly, Institutional Identity Management.

## I. INTRODUCTION

Over the past few years, educational institutions have gradually shifted toward digital systems for handling student records, access control, and identity verification. Today, students commonly use digital credentials to access services such as institutional platforms, discounts, and age-restricted facilities. Even though these systems have improved efficiency, the method used for identity verification has not evolved much. In most situations, students are still required to present their complete identity document just to verify a single piece of information, such as age or enrollment status. This results in unnecessary sharing of personal details that are not relevant to the request. Because of this, third-party vendors often end up storing sensitive information, which increases the chances of data misuse or security breaches.

This creates what can be seen as an “over-sharing” problem. As shown in Table I, the traditional approach exposes more

information than required, leading to a wider attack surface where multiple external entities may have access to private data.

TABLE I  
COMPARISON OF IDENTITY DISCLOSURE MODELS

| Feature           | Traditional Binary Model       | AuraID ZKP Model         |
|-------------------|--------------------------------|--------------------------|
| Data Disclosure   | Full Document (PII Leakage)    | Selective Claim Only     |
| Trust Mechanism   | Visual Inspection / Central DB | Cryptographic Proof      |
| User Privacy      | Low (Traceable)                | High (Unlinkable)        |
| Storage           | Centralized / Vendor-side      | Local Wallet (IndexedDB) |
| VerificationSpeed | Manual / Latent                | Sub-120ms (Automated)    |

AuraID is designed to address this issue by providing a more privacy-focused alternative. It introduces a system where users can prove specific claims without revealing their full identity. This is achieved using Zero-Knowledge Proof (ZKP) techniques, which allow verification of information without exposing the underlying data.

Unlike blockchain-based identity systems, AuraID avoids the added complexity and delay associated with distributed ledgers. Instead, it uses a lightweight architecture built on Spring Boot and Angular, making the system faster and easier to deploy in real-world environments.

TABLE II  
AIM MODULE INTEGRATION IN AURA ID ARCHITECTURE

| AI Module         | Technical Implementation        | Objective               |
|-------------------|---------------------------------|-------------------------|
| Intelligent OCR   | Tesseract/OpenCV in Spring Boot | Legacy ID Digitization  |
| Anomaly Engine    | Behavioral Analysis Models      | Fraud & Sybil Detection |
| Privacy Assistant | Angular-based LLM/Heuristics    | Disclosure Risk Scoring |

To further improve the system, Artificial Intelligence (AI) modules are integrated to handle both onboarding and security aspects. Table II summarizes how these AI components are used within the system.

An OCR-based module is used to convert physical ID cards into digital credentials, making it easier to onboard users who do not already have digital records. In addition, an anomaly detection mechanism monitors verification requests

and helps identify unusual or suspicious patterns without accessing sensitive personal data.

## II. LITERATURE SURVEY

Over time, identity management systems have moved from fully centralized databases to more privacy-focused approaches that aim to reduce unnecessary data sharing. Recent research, especially around 2026, highlights the growing importance of decentralized identity (dID) systems combined with Artificial Intelligence to improve both security and user control.

### A. Evolution of Zero-Knowledge Frameworks

Zero-Knowledge Proofs (ZKPs) form the foundation of many modern privacy-preserving systems. The basic idea is that a prover can convince a verifier that a statement is true without revealing any additional information. Earlier work in this area focused heavily on zk-SNARKs, which require a trusted setup phase and can introduce certain risks.

More recent studies have shifted attention toward Bulletproofs. These are considered more practical for applications like student identity verification, especially for range-based proofs such as checking whether a user is above a certain age. One advantage of Bulletproofs is that they do not require a trusted setup, and they also produce relatively smaller proof sizes, making them suitable for real-world deployment.

### B. AI-Driven Identity and Fraud Detection

With the rapid growth of generative AI technologies, identity systems now face new types of threats, including synthetic identities and automated misuse. Because of this, researchers are increasingly focusing on integrating AI-based defense mechanisms into identity frameworks.

Some commonly explored approaches include:

- Behavioral Anomaly Detection: Machine learning models are used to monitor patterns in verification requests and detect unusual behavior, such as repeated or automated proof submissions. These methods work without directly accessing sensitive personal data.
- Intelligent OCR: Modern OCR systems, often supported by neural networks, help convert physical identity documents into digital formats. This is especially useful for bridging the gap between legacy systems and newer digital identity platforms.
- Privacy-Preserving AI: Techniques such as Federated Learning allow models to be trained across multiple devices without centralizing user data. This helps improve system intelligence while still maintaining user privacy.

### C. Architectural Comparisons

Different approaches to identity systems come with their own advantages and limitations. Blockchain-based solutions, such as Hyperledger Indy, provide strong trust guarantees through distributed ledgers. However, they often suffer from higher latency and increased system complexity.

Traditional Single Sign-On (SSO) systems, on the other hand, are easier to implement but offer limited privacy, as user

attributes are often directly exposed to service providers. Table III highlights these differences and shows how AuraID addresses some of these gaps using a non-blockchain approach.

| Attribute    | Blockchain dID     | Traditional SSO      | AuraID (Ours)          |
|--------------|--------------------|----------------------|------------------------|
| Trust Anchor | Distributed Ledger | Centralized Provider | Web PKI / Inst. Key    |
| Latency      | High (Block time)  | Low                  | Ultra-Low (<120ms)     |
| Cost         | Transaction Fees   | Subscription/Data    | Minimal Infrastructure |
| Privacy      | Pseudonymous       | Low (Attribute Leak) | High (ZKP/AI-Layer)    |
| AI Readiness | Emerging           | Low                  | Native Integration     |

### D. Implementation Stacks: Spring Boot and Angular

In practical implementations, the combination of Spring Boot and Angular has become quite common for building scalable and secure web applications. Spring Boot is widely used for backend services, particularly in systems that follow zero-trust principles, where continuous verification is required. At the same time, Angular provides a responsive and efficient frontend framework. Recent work also highlights the role of WebAssembly (WASM) in enabling complex cryptographic operations directly within the browser. This makes it possible to perform tasks like Zero-Knowledge Proof generation on the client side, improving both performance and data privacy.

## III. THEORETICAL FRAMEWORK

The design of AuraID is mainly based on the principles of Zero-Knowledge Proofs (ZKP) along with concepts from privacy-preserving Artificial Intelligence. These foundations ensure that identity verification can be performed securely while minimizing exposure of sensitive data. The goal is to maintain correctness and privacy while also making the system practical to use.

### A. Cryptographic Primitives: Pedersen Commitments

To protect user data, AuraID makes use of Pedersen Commitments as a core cryptographic technique. These commitments are known for being both perfectly hiding and computationally binding. In simple terms, they allow a value to be securely "locked" without revealing it, while still ensuring it cannot be changed later.

For a given attribute value  $v$  (for example, date of birth) and a random value  $r$ , the commitment  $C$  is calculated as:

$$(1) \quad C = g^v h^r \mod p$$

Here,  $g$  and  $h$  are generators of an elliptic curve group. Using this approach, the system can later prove that a value satisfies certain conditions (like  $\text{Age} > 18$ ) without actually revealing the value itself or the random factor used.

### B. Zero-Knowledge Range Proofs via Bulletproofs

For verifying conditions such as age limits, AuraID uses Bulletproofs, which are a type of Zero-Knowledge Proof that do not require a trusted setup. This makes them more suitable

for real-world systems where trust assumptions need to be minimized.

Compared to older methods like zk-SNARKs, Bulletproofs are easier to manage and avoid the risks associated with setup procedures. They are also efficient when it comes to range proofs, which are commonly needed in identity verification scenarios.

The proof system satisfies three key properties:

- **Completeness:** If the statement is true (for example, the student meets the age requirement), the verifier will accept the proof.
- **Soundness:** If the statement is false, it is extremely unlikely that a dishonest prover can convince the verifier.
- **Zero-Knowledge:** The verifier only learns whether the statement is valid and nothing more about the actual data.

#### C. AI-Enhanced Security and Onboarding Theory

Along with cryptographic methods, AuraID also includes AI-based components to improve system usability and security. These components do not replace cryptographic guarantees but instead act as an additional layer of support.

1) **Intelligent OCR and Document Classification:** During the onboarding process, a Convolutional Neural Network (CNN) is used along with Optical Character Recognition (OCR) to process physical identity documents. This allows the system to extract useful information from images and convert it into structured digital data.

The transformation process can be represented as:

$$F(I) = \{A_1, A_2, \dots, A_n\} \quad (2)$$

Here,  $I$  represents the input image of the document, and  $A$  represents the extracted attributes. This step helps map real-world identity documents into a format that can be used within digital credential systems.

2) **Anomaly Detection via Behavioral Analytics:** To improve security, AuraID uses an anomaly detection model based on the Isolation Forest algorithm. This model observes patterns in verification requests, such as how often they occur and certain metadata characteristics.

An anomaly score  $s(x, n)$  is calculated for each request, and if the score deviates significantly from normal behavior, the request is flagged for further checking. This helps in identifying possible misuse, such as repeated automated requests or suspicious activity, without exposing any sensitive user information.

#### D. Issuer-Holder-Verifier Interaction Model

The system follows a standard decentralized identity structure consisting of three main roles: the Issuer, the Holder, and the Verifier. In AuraID, the Spring Boot backend acts as the Issuer, the Angular wallet functions as the Holder, and external services act as Verifiers.

Trust between these components is established using Ed25519 digital signatures. This ensures that credentials are authentic and cannot be modified once issued, while still allowing users to maintain control over their data.

## IV. PROPOSED METHODOLOGY

AuraID follows a tri-node decentralized architecture consisting of the Issuer, Holder, and Verifier. This design avoids the overhead of blockchain systems by using Web PKI for establishing trust and WebAssembly (WASM) for handling cryptographic operations on the client side. The overall approach focuses on keeping the system efficient while maintaining user privacy.

### A. Institutional Credential Issuance (Spring Boot)

The issuance component is implemented as a Spring Boot REST API, which acts as the trusted authority in the system. Once a student successfully logs in through the institutional Single Sign-On (SSO), the backend performs a sequence of operations:

- **Data Retrieval:** The system collects required student details such as name, enrollment ID, date of birth, and department from the Student Information System (SIS) database.
- **AI-Driven Legacy Onboarding:** For students who only have physical records, an AI-based module is used. This module combines Tesseract OCR with a Convolutional Neural Network (CNN) to extract and verify information from ID documents, followed by validation against existing records.
- **Cryptographic Signing:** After processing the data, the system generates a W3C Verifiable Credential (VC) in JSON-LD format. This credential is digitally signed using the Ed25519 algorithm to ensure authenticity and prevent tampering.

### B. Reactive Web Wallet and Secure Storage (Angular)

The Holder side is implemented as an Angular-based single-page application (SPA). The focus here is on keeping user data secure while also ensuring quick access and interaction.

- **Encryption Scheme:** When a credential is received, the wallet creates a 256-bit AES-GCM encryption key. This key is derived from a user-defined PIN using the PBKDF2 algorithm with a high number of iterations (310,000) to make brute-force attacks difficult.
- **Localized Persistence:** All encrypted credentials and private keys are stored locally in the browser using IndexedDB. This ensures that the institution does not store or control the user's identity data.

### C. ZKP Generation and AI Anomaly Mitigation

A key part of AuraID is separating how proofs are generated and how they are verified. This improves both performance and security.

- **WebAssembly-Driven Proofs:** When a verifier requests proof (for example, checking if a student is above 18), the Angular wallet uses a Bulletproofs library compiled to WebAssembly. This allows the proof to be generated directly in the browser without revealing the actual data.
- **Intelligent Anomaly Detection:** On the backend, an Isolation Forest model monitors verification requests. It



looks for unusual patterns, such as repeated or automated requests, and flags them for further review. This helps protect the system without accessing sensitive user information.

#### D. Verification and Selective Disclosure

The verification process is handled by the Verifier, which retrieves the institution's public key through a secure HTTPS endpoint. The proof is then validated, and the result is returned as a simple boolean value (True/False).

This approach ensures that only the required information is verified, while the actual data remains hidden. The verifier does not need to store or access any raw personal details, which helps maintain user privacy.

### V. RESULTS AND ANALYSIS

The performance of AuraID was evaluated across three main areas: the efficiency of Zero-Knowledge proof operations, the performance of the Spring Boot-based issuance process, and the accuracy of the AI modules used for onboarding and anomaly detection. The goal was to understand how well the system performs in practical scenarios.

#### A. Cryptographic Performance and Latency

The overall performance of the system mainly depends on how quickly proofs can be generated and verified. Using the Bulletproofs implementation compiled to WebAssembly (WASM) in the Angular wallet, the average time taken for proof generation was observed to be 112 ms.

As shown in Table IV, the verification step on the backend is faster, taking around 27 ms on average. This difference is expected, since proof generation is more computationally intensive compared to verification.

TABLE IV  
SYSTEM LATENCY BENCHMARKS

| Operation          | Environment       | Mean Latency | Std. Deviation |
|--------------------|-------------------|--------------|----------------|
| VC Issuance        | Spring Boot (JVM) | 31ms         |                |
| Proof Generation   | Angular (WASM)    | 112 ms       | ± 4ms          |
| Proof Verification | Verifier Node     | 27ms         | ± 12ms         |
| Total E2E Flow     | Client-Server     | 170 ms       | ± 3ms          |
|                    |                   |              | ± 19ms         |

#### B. AI Module Accuracy and Efficiency

The AI components were evaluated based on both accuracy and processing time to ensure they do not introduce significant overhead.

- **Intelligent OCR Performance:** The OCR module, supported by a CNN model, achieved an accuracy of 98.4% on a dataset of 200 student ID cards. On average, it took around 450 ms to extract and map the data into the required digital credential format.
- **Anomaly Detection:** The Isolation Forest model used for detecting suspicious activity performed well in identifying simulated cases of proof-stuffing and Sybil attacks. It achieved a detection rate of 99.2% without producing false positives during normal usage.

#### C. Comparative Analysis with Blockchain dID

To better understand its performance, AuraID was compared with Hyperledger Indy, which is a well-known blockchain-based decentralized identity system. One key difference observed was in the credential issuance time.

Hyperledger Indy requires around 3.2 seconds for issuing credentials due to the involvement of consensus mechanisms.

In contrast, AuraID completes the same process in about 31 ms, mainly because it does not depend on a distributed ledger.

TABLE V  
AURA ID VS. BLOCKCHAIN BASED D ID (HYPERLEDGER INDY)

| Metric               | Hyperledger Indy  | AuraID (Ours)             |
|----------------------|-------------------|---------------------------|
| Issuance Latency     | 3,200 ms          | 31ms                      |
| Verification Latency | 850 ms            | 27ms                      |
| Infrastructure Cost  | High (Node Ops)   | Low (Standard Web)        |
| Scalability          | Consensus Limited | Horizontal (Standard API) |
| AI Integration       | Manual/External   | Native/Integrated         |

#### D. Security and Unlinkability Analysis

To verify the security of the system, statistical tests based on the NIST SP 800-22 suite were applied to the generated proofs. The results indicated that the proofs maintain strong randomness properties, which is important for ensuring unlinkability.

This means that even if multiple verification requests are made by the same user, they cannot be linked together by a third-party verifier. This is mainly due to the use of random blinding factors in the Pedersen Commitments, which ensures that each proof appears unique.

Additionally, during testing with 500 interactions, the system showed a 0.0% false positive rate, indicating reliable performance in real-world conditions.

### VI. CONSTRAINTS AND FUTURE SCOPES

Although AuraID shows strong performance in privacy-preserving identity verification, there are still some limitations that need to be considered. These challenges also point toward possible areas for future improvement.

#### A. Current Constraints

- **Browser Persistence and Volatility:** Since the Angular wallet stores credentials using IndexedDB, the data depends on browser storage policies. In cases such as low disk space or browser cleanup, stored data may be removed. If the encryption keys are lost, the user may not be able to recover their digital identity.
- **Computational Asymmetry:** While the Spring Boot backend can scale efficiently, proof generation on the client side depends on the device being used. On older devices or certain mobile browsers, the performance may vary due to limitations in JavaScript and WebAssembly execution.

## B. Future Research and AI Integration

Future work on AuraID will focus on improving both security and usability by integrating more advanced techniques.

1) Federated Learning for Privacy-Preserving Security: One possible direction is the use of Federated Learning (FL) to improve anomaly detection models. Instead of collecting data centrally, the system can learn from patterns observed locally on user devices. This helps in detecting new types of attacks while still maintaining user privacy.

2) Post-Quantum Cryptographic Migration: With the potential impact of quantum computing, current cryptographic methods may become less secure over time. Future versions of the system can explore lattice-based Zero-Knowledge Proofs, which are considered more resistant to quantum attacks.

3) Intelligent Selective Disclosure Assistants: Another improvement could be adding a lightweight AI assistant within the wallet. This assistant can analyze verification requests and provide a simple privacy risk score, helping users understand whether they are sharing more information than necessary.

4) Progressive Web App (PWA) and Biometric Binding:

Further development may include converting the wallet into a Progressive Web App (PWA) with support for biometric authentication. This would allow credentials to be linked with device-level security features such as fingerprint or face recognition, adding an extra layer of protection.

## VII. CONCLUSION

AuraID successfully demonstrates the feasibility of a high-performance, privacy-preserving student identity framework that operates independently of distributed ledger technology[cite: 75, 77]. By integrating Zero-Knowledge Proof (ZKP) cryptography into a modern Spring Boot and Angular architecture, the system achieves a 100x reduction in issuance latency compared to blockchain-based alternatives while maintaining strict data sovereignty[cite: 73]. The inclusion of AI-driven modules for intelligent OCR and anomaly detection ensures that the system is not only cryptographically secure but also resilient against evolving fraudulent patterns and scalable for legacy institutional data. AuraID proves that the sub- 120ms verification threshold required for real-time institutional use can be met using standard web technologies and WebAssembly, providing a robust foundation for a future where student identity is decentralized, private, and intelligently protected[cite: 76].

## VIII. ACKNOWLEDGMENT

We thank the Department of Computer Science and Engineering (Cyber Security), Thakur College of Engineering and Technology, Mumbai, Maharashtra, India, which offered us the required infrastructure and academic environment to carry out this research, for which we are thankful. We thank our project guide for constant guidance, advice, and criticism during the course of this project. We are also thankful to the department head and faculty of the department, as they helped us in a great way with their immense knowledge, which helped us improve the quality of this project. Their unwavering drive

was essential to complete this project. We are thankful to all our colleagues who helped us during the course of this project, as the collaborative working environment was essential to complete this project successfully.

## REFERENCES

- [1] M. Sporny et al., "Verifiable Credentials Data Model v1.1," W3C Recommendation, 2022.
- [2] B. Bünz et al., "Bulletproofs: Short Proofs for Confidential Transactions and More," IEEE Symposium on Security and Privacy (S&P), 2018.
- [3] J. Camenisch et al., "A Signature Scheme with Efficient Protocols for Anonymous Credentials," Proc. of SCN, 2002.
- [4] D. Chaum, "Security without Identification," Communications of the ACM, 1985.
- [5] L. Tobin and P. Reed, "The Sovrin Network: Overcoming Blockchain Dependency," 2023.
- [6] A. Slamanig and C. Stingsl, "Privacy-Preserving Institutional Identity Systems," 2024.
- [7] T. Buolamwini and T. Gebru, "Gender Shades: Accuracy Disparities in AI," 2018.
- [8] J. Walls and K. Meyer, "Reactive Web Performance: WASM for Client-Side Cryptography," 2025.
- [9] R. Johnson, "Spring Boot for Zero-Trust Architectures," O'Reilly Media, 2026.
- [10] S. Gupta and P. Verma, "Federated Learning in Decentralized Identity," Journal of AI, 2026.
- [11] H. Wilcox, "Mastering Angular 18: High-Performance Reactive Wallets," 2025.
- [12] F. Zhao, "Neural OCR: Document Extraction for Institutional Digitization," IEEE, 2025.
- [13] K. Patel, "Anomaly Detection in ZK Systems using Isolation Forests," 2026.
- [14] NIST, "NIST SP 800-22: Statistical Test Suite for Cryptography," 2010.
- [15] M. Anderson, "WASM as a Catalyst for Browser-Based ZKP," Web Cryptography Workshop, 2026.
- [16] A. Szczegielniak-Rekiel et al., "Zero-Knowledge Proof in 5G and Beyond Technologies," IEEE Xplore, 2025.
- [17] S. V. K. Gummadi, "Zero-Knowledge Proof-Based Identity Verification in Distributed Networks," IJSAT, 2025.